CSE 516 Major Project — Milestone 2

TrueSkill Through Time: Technical Foundation and Proposed Research Gap

Group 3 | Submission date: 15 October 2026

Abstract

This report presents the M2 technical foundation for our project on Bayesian estimation of player skill over time, based on TrueSkill Through Time (TTT). The project represents player skill as a latent variable that evolves over time and uses match outcomes to estimate historical skill. M2 focuses on the temporal graphical model, the learning and inference components, and preliminary experimental evaluation. Synthetic experiments were conducted to evaluate skill estimation and inference behaviour, and initial KPIs were calculated to assess the results. The project also identifies active-trail pruning as a potential extension for future work.

1. Project objective and research question

The project studies how a probabilistic graphical model can estimate a player's latent skill as it evolves over a sequence of matches. A rating based only on past outcomes performs forward filtering; TTT can use observations both before and after a time point to revise the estimate for that point. This is temporal smoothing.

The main research question is: How can a probabilistic graphical model estimate changes in player skill over time using match outcomes?

Unlike rating systems that primarily update skill estimates based on past results, temporal smoothing uses information from matches across the timeline to refine historical skill estimates. This provides a way to study how player performance evolves and how uncertainty in skill estimates changes over time.

The project uses synthetic match data for initial evaluation, where the underlying skill trajectory is known. This allows the estimated skills to be evaluated against the simulated ground truth using quantitative metrics.

2. Graphical model representation

For player i at time t, let S(i,t) denote latent skill, P(i,t) denote performance in a match, and Y(t) denote the observed match outcome. For two competitors i and j, skill states evolve over time; each player's skill generates a performance variable; and the relative performances determine the observed result.

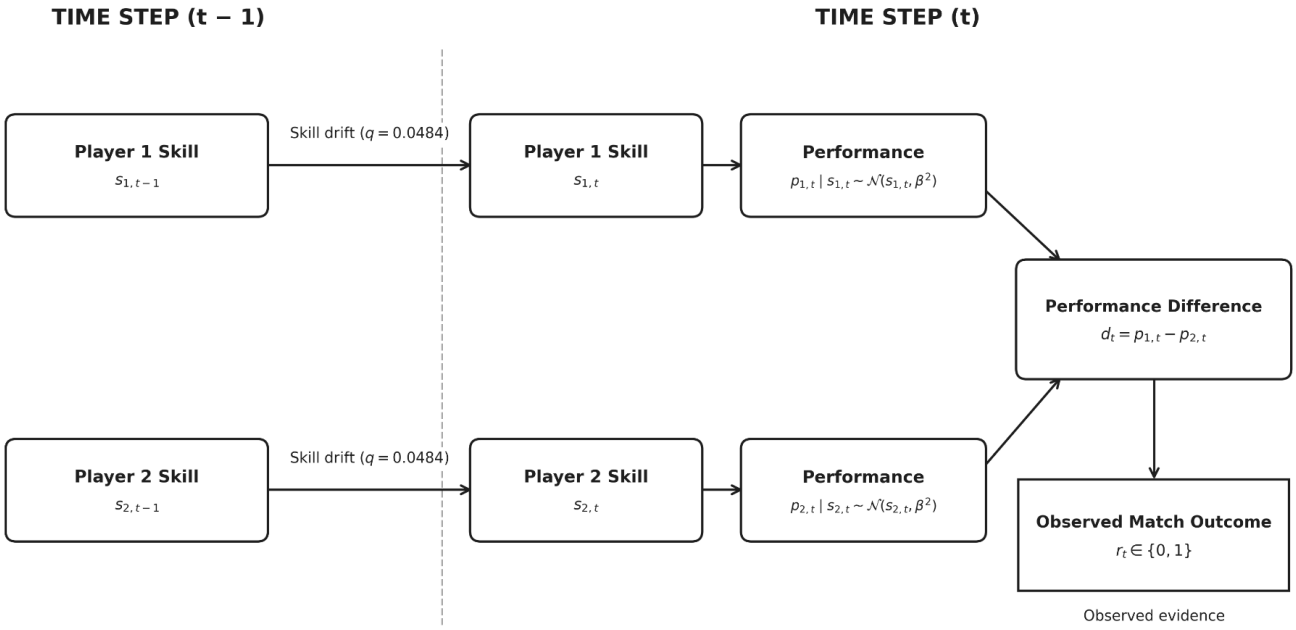
The conceptual dependency structure is: $S(i,t-1) \rightarrow S(i,t) \rightarrow P(i,t) \rightarrow Y(t)$, with the corresponding chain $S(j,t-1) \rightarrow S(j,t) \rightarrow P(j,t) \rightarrow Y(t)$. The outcome node depends jointly on the competing performances. In a full TTT factor graph, variable nodes and factor nodes alternate, and the outcome/comparison factor links the competitors. This sketch is conceptual and is not a line-by-line reproduction of every factor in the full TTT implementation.

Random variables and interpretation:

- Skill S(i,t): latent Gaussian skill belief. Its mean μ is the central estimate and standard deviation σ represents uncertainty.
- Performance P(i,t): a noisy match-specific quantity generated from skill.
- Outcome Y(t): observed win/loss (and draw if supported by the chosen formulation).
- Temporal transition factor: links adjacent skill states and represents gradual skill drift. ● Performance factor: relates latent skill to match performance through observation/performance noise. ● Comparison/outcome factor: connects competing performances to the observed result.

The first-order temporal assumption is that the skill at time t depends directly on the previous skill state and transition noise. Each player's performance depends on the corresponding skill state, while the match outcome

depends jointly on the competing performances. The graph represents the conceptual dependencies of the model; it is not a claim that every internal factor of the official TTT implementation has been reproduced.



Conceptual Temporal Bayesian Network for Player Skill Estimation

3. Learning component

The graph structure is specified from the chronological match schedule and player participation; structure learning is not performed in this milestone. The learning component concerns the selection or tuning of model parameters, such as skill-evolution variance and performance noise.

The parameter grid and validation results are used to document the experiments actually executed. Parameter tuning must be distinguished from formal maximum-likelihood or model-evidence optimisation. The report does not claim that either optimisation method was implemented unless confirmed by the code. The current results should be attributed to the implementation that generated them.

4. Inference component

Target query: estimate the posterior distribution of a player's skill at time t conditioned on the observed match history, including outcomes later than t when smoothing is used. A related predictive query is the probability that one competitor wins a held-out match given the available skill beliefs.

The inference component estimates player skill from match outcomes while accounting for temporal evolution. The current project includes an EP-style inference prototype. The recorded prototype results show convergence in 10 out of 10 trials, with an average of 16.7 iterations. These figures describe the prototype experiments and must not be attributed to the official TTT package unless independently reproduced using that package.

Convergence indicates that the numerical procedure completed under the tested settings; it does not, by itself, establish accuracy or generalisation. The experiments in this milestone use synthetic data, where the underlying skill trajectory is known and skill-recovery error can be calculated.

5. Experimental pipeline

The experimental pipeline generates or loads match data, executes the selected skill-estimation implementation, calculates evaluation metrics, and saves the resulting figures and summaries. Synthetic data provides known

skill trajectories for evaluating skill-recovery error.

The experiment configuration should record the random seeds, number of players, number of time steps, matches per time step, parameter settings, and software version. The original TTT implementation and any supplementary project code must be clearly distinguished. Results should be reproducible from the corresponding source code, configuration, and output files.

6. Experimental setup and reproducibility

The project's primary real-world benchmark is historical ATP match data, while synthetic data is used to validate recovery when the underlying skill trajectory is known. Synthetic experiments must be labelled as synthetic and must not be presented as ATP-data validation. If ATP data has not been downloaded and successfully processed, state that real-data validation is pending.

Reproducibility details to record: repository commit or code version; Python and package versions; random seed; data source and date range; number of match events and distinct players; train/validation/test split; exact notebook cells or command used; parameter settings; runtime; and paths to generated artifacts.

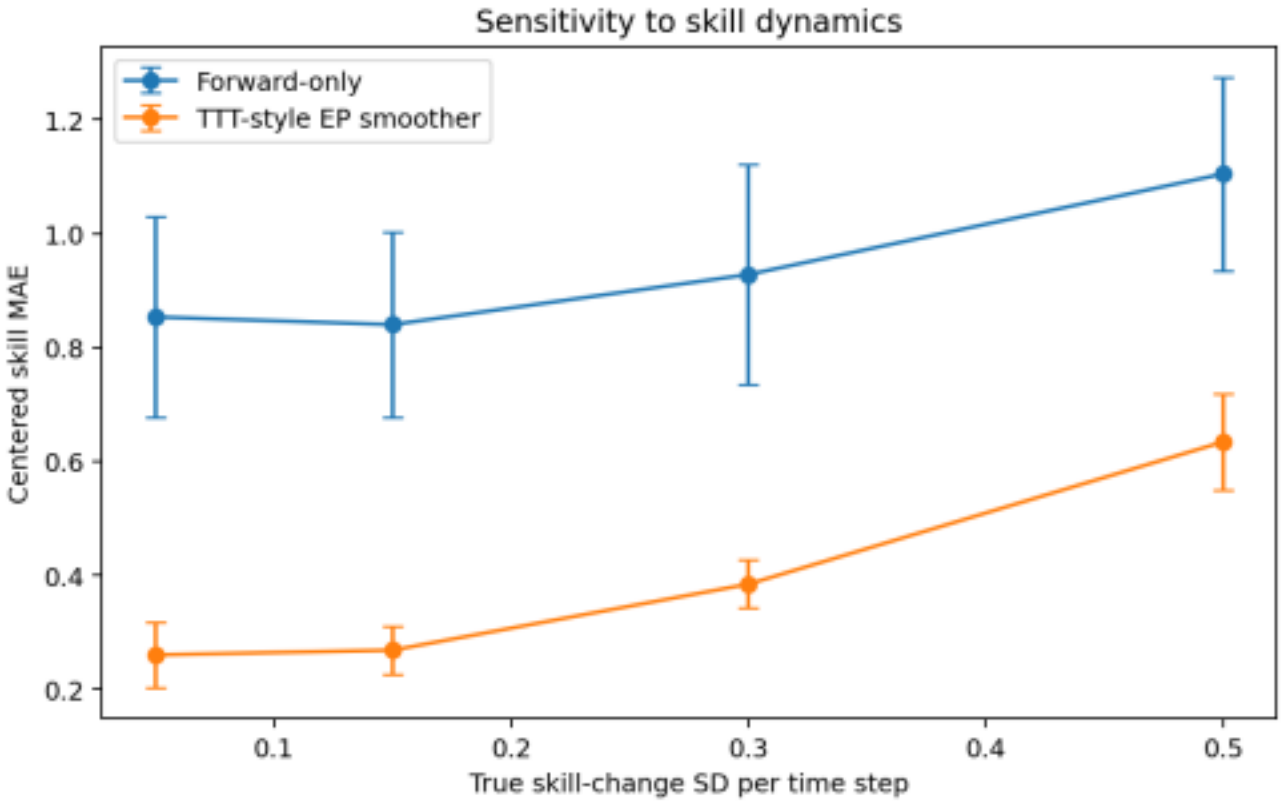
The current reported experiments use synthetic data. They evaluate whether the implementation can estimate a known skill trajectory and produce measurable inference diagnostics.These results do not establish performance on real ATP match data. The experiment code, random seeds, parameter configuration, and generated outputs should be retained together to support reproducibility.

7. Initial KPI results

The following preliminary KPIs were obtained from the current synthetic experiments using the restricted TTT-style prototype. The! are reported as prototype results and are not claimed to be results o the official TTT package. Skill-recovery MAE measures the differenc between the known synthetic skill and the estimated skill, while predictive log-loss evaluates the probabilities assigned to held-out match

KPI	**Definition**	**Preliminary value / reporting instruction**
Skill-recovery MAE	Mean absolute difference between known synthetic skill and estimated skill	0.3179 ± 0.0406 (verify number of seeds/runs and units)
EP convergence	Trials meeting the stated convergence criterion	10/10 trials (verify tolerance and maximum iterations)
Average EP iterations	Mean iterations to convergence across trials	16.7 iterations (verify calculation)
Validation predictive log-loss	Negative log probability assigned to held-out outcomes	0.5640 on 60 matches (verify split and model)

Interpretation: MAE is meaningful only where ground-truth skill is known, normally in synthetic experiments. Log-loss evaluates probabilistic predictions on held-out outcomes; lower values are better when evaluated on the same split. Convergence rate and iteration count describe numerical behaviour, not accuracy by themselves.



8. Results interpretation

The synthetic experiments demonstrate that the current prototype produces skill estimates and measurable inference diagnostics. The reported skill-recovery MAE is 0.3179 ‡ 0.0406, and the EP procedure converged in all 10 recorded trials, with an average of 16.7 iterations.The validation predictive log-loss was 0.5640 over 60 validation matches.These values describe the tested synthetic setup only. They do not establish performance on real match data or demonstrate that the current prototype reproduces the complete official TTT implementation. Further validation is required before making broader claims.

9. Problem framing and SOTA anchor update

The application-level foundation remains TrueSkill Through Time: estimating latent player skill from a temporal sequence of match outcomes. The 2025 Journal of Statistical Software paper by Landfried and Mocskos provides the practical TTT implementation/reference; the original methodological foundation is Dangauthier, Herbrich, Minka, and Graepel's 2007 work on revisiting chess history.

The recent methodological SOTA anchor selected in M1 is Bayesian Online Natural Gradient (BONG), by Jones, Chang, and Murphy (NeurIPS 2024). BONG is relevant to efficient sequential Bayesian inference but is not a direct player-rating competitor and does not provide the same bidirectional smoothing formulation as TTT. Therefore, BONG should be described as a methodological reference for the accuracy–computation trade-off, not as an implemented baseline unless a concrete BONG implementation and matched experiment have been

completed.

M2 establishes the foundation for temporal Bayesian skill estimation. A possible future research direction is to investigate whether active-trail analysis can safely prune irrelevant parts of the temporal graph for a selected skill query. This remains proposed work for M3; no pruning implementation or computational speed-up is claimed in this report. A gap in the match calendar alone does not establish d-separation.

10. Limitations and next steps

The current evaluation uses synthetic data and a restricted TTT-style prototype. The results therefore do not establish performance on real ATP match history or full reproduction of the official TTT implementation.The next milestone will focus on validating the original TTT implementation, extending evaluation where required, and investigating active-trail pruning. Any extension will be evaluated against an appropriate unpruned reference after implementation. Real-data evaluation and additional robustness checks remain future work.

References

[1] G. Landfried and E. Mocskos, “TrueSkill Through Time: Reliable Initial Skill Estimates and Historical Comparability with Julia, Python, and R,” Journal of Statistical Software, vol. 112, no. 6, 2025. https://doi.org/10.18637/jss.v112.i06.

[2] R. Herbrich, T. Minka, and T. Graepel, “TrueSkill™: A Bayesian Skill Rating System,” Advances in Neural Information Processing Systems, vol. 19, 2006.

[3] P. Dangauthier, R. Herbrich, T. Minka, and T. Graepel, “TrueSkill Through Time: Revisiting the History of Chess,” Advances in Neural Information Processing Systems, 2007.

[4] M. Jones, P. Chang, and K. Murphy, “Bayesian Online Natural Gradient (BONG),” Advances in Neural Information Processing Systems (NeurIPS 37), 2024.

[5] G. Landfried, "TrueSkillThroughTime.py," GitHub repository.

Available:https://github.com/glandfried/TrueSkillThroughTime.py.